

PHYS 206 Experiment 1 Lab Report: Geometric Optics

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Part A: Converging Lens – Image and Object Relationship

1 Introduction

Given a lens of any shape and index of refraction, you could determine the shape and location of the images it forms based only on the Law of Refraction. You need only apply the law along with some of the ray tracing techniques you have already used.

However, for spherical lenses (and for spherical mirrors as well), there is a more general equation that can be used to determine the location and magnification of an image. This equation is called the Fundamental Lens Equation.

In this experiment, we test and apply the relationship between the object distance, image distance, and focal length of a converging lens.

2 Theory

Fundamental Lens Equation

For a thin spherical lens, the relationship between the focal length (f), the object distance (d_o), and the image distance (d_i) is

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \quad (1)$$

where f is the focal length of the lens, d_o is the distance from the lens to the object and d_i is the distance from the lens to the image

The magnification of the image is given by

$$m = -\frac{d_i}{d_o} \quad (2)$$

Instead of the above equation, we may write the Fundamental Lens Equation as

$$S_o S_i = f^2 \quad (3)$$

where S_o and S_i are the distances between the principle focus of the lens and the object and image, respectively. If so, notice that

$$S_o = d_o - f \quad S_i = d_i - f$$

These equations allow us to determine both the position and the size of the image formed by a converging lens. See Figure 1 for physical representations of the variables.

3 Experimental Setup

Equipment

The following equipment was used:

- Optics Bench
- Light Source
- 75 mm Focal Length Convex Lens
- Crossed Arrow Target
- Component Holders
- Viewing Screen

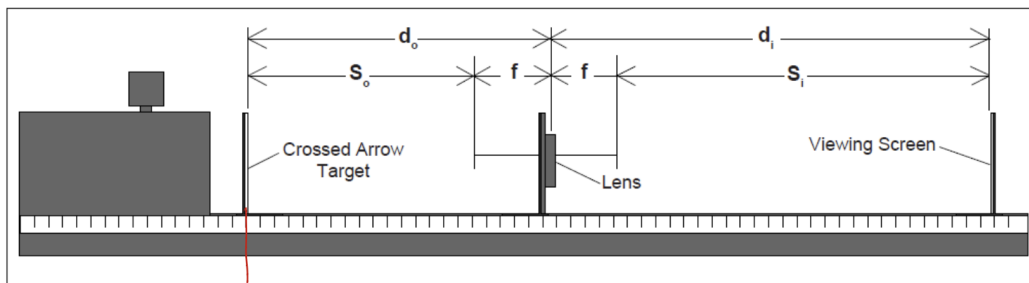


Figure 1: Equipment Setup

Procedure

1. Set up the equipment on the optics bench as shown in Figure 1.
2. Turn on the Light Source and slide the lens toward or away from the Crossed Arrow Target, as needed to focus the image of the Target onto the Viewing Screen

4 Data Analysis

Table 1: Optics Data and Calculations

Data			Calculations			
d_o (mm)	d_i (mm)	h_i (mm)	$1/d_i + 1/d_o$	$1/f$	h_i/h_o	$-d_i/d_o$
500	90	1	0.0131	0.0133	0.05	-0.18
450	100	4	0.0122	0.0133	0.20	-0.22
400	110	5	0.0116	0.0133	0.25	-0.28
350	100	4	0.0129	0.0133	0.20	-0.29
300	100	8	0.0133	0.0133	0.40	-0.33
250	115	6	0.0127	0.0133	0.30	-0.46
200	130	10	0.0127	0.0133	0.50	-0.65
150	140	16	0.0138	0.0133	0.80	-0.93
100	180	22	0.0156	0.0133	1.10	-1.80
75	210	26*	0.0181	0.0133	1.30	-2.80
50	190	20*	0.0253	0.0133	1.00	-3.80

*: Not clear enough.

Even if we try to measure h_i when d_o is smaller than the focal length, these measurements are not reasonable because the image is imaginary at these distances. Thus, I will not use these two values during calculations.

By the Fundamental Lens Equation

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

The average focal length was calculated using the first nine data points (where $d_o > f$). The experimental focal length f_{exp} is given by:

$$f_{exp} = \frac{1}{n} \sum_{i=1}^n f_i = 77.4 \text{ mm}$$

The percentage error relative to the theoretical focal length ($f_{th} = 75 \text{ mm}$) is calculated as:

$$\text{Error } \% = \left| \frac{f_{th} - f_{exp}}{f_{th}} \right| \times 100 = \%3.2$$

To evaluate the overall accuracy of the magnification measurements, the average percent error ($\bar{E}$) was calculated by comparing the theoretical magnification ($m_{th} = -d_i/d_o$) and experimental magnification ($m_{exp} = h_i/h_o$) for each data point:

$$E_i = \left| \frac{m_{th,i} - m_{exp,i}}{m_{th,i}} \right| \times 100$$

$$\bar{E} = \frac{1}{n} \sum_{i=1}^n E_i$$

The average percent error for magnification was found to be approximately %15.4 (excluding values where $d_o < f$).

5 Discussion

5.1 Questions

1. Is the image magnified or reduced?

Answer: We can write $m = \frac{f}{f-d_o}$ using the magnification equation and the Fundamental Lens Equation. While we can observe the image ($d_o > 75$ mm), we expect the image would be always smaller than the object. Our results except $d_o = 100$ mm satisfied that. When $d_o = 100$ mm it was magnified and it might be caused by our measurement errors because while we keep d_i bigger, it was getting harder to observe the clear image. In conclusion, it should be **reduced** and our results mostly satisfied it.

2. Is the image inverted?

Answer: Yes, it is inverted. This is because the - sign in Equation 2 and we observed it experimentally.

3. Based on the Fundamental Lens Equation, what would happen to d_i if d_o were increased even further?

Answer: When we leave only d_i in the Fundamental Lens Equation, it will be $d_i = \frac{d_o f}{d_o - f}$. f is a constant depending on the geometry of the

lens, which is 75mm, so $\lim_{d_o \rightarrow \infty} d_i$ would decrease and converge 75mm if we increased d_o to infinity.

4. What would happen to d_i if d_o were very large?

Answer: It converges to 75mm (f) as I said in previous question.

5. Using your answer to the previous question, estimate the focal length of the lens.

Answer: 75mm

6. Are your results in complete agreement with the Fundamental Lens Equation? If not, explain possible sources of discrepancy.

Answer: Our results are very near to theoretical values and the error is %3.2 for focal length as I showed in Data Analysis section. Our data is a bit more mistaken for magnification which is %15.4 and it might be caused because of our measurement errors and misunderstanding some part of the experiment procedure.

7. For what values of d_o were you unable to focus an image onto the screen? Use the Fundamental Lens Equation to explain why.

Answer: As I mentioned in Data Analysis section, when $d_o < f$, our observations are not clear and I do not use them while calculations and answering questions because the image is virtual in this range.

5.2 Additional Questions

1. For a lens of focal length f , what value of d_o would produce an image with a magnification of one?

Answer: $|m| = 1$ if $d_i = d_o$. Substituting this into the Fundamental Lens Equation:

$$\frac{1}{d_o} + \frac{1}{d_o} = \frac{1}{f} \implies \frac{2}{d_o} = \frac{1}{f} \implies d_o = 2f$$

2. Is it possible to obtain a non-inverted image with a converging spherical lens? Explain.

Answer: Yes, it is possible. When the object is between the focal

point and the lens ($d_o < f$), the lens produces a virtual and non-inverted image. However, since the image is virtual, it cannot be projected onto a screen and must be viewed through the lens.

3. For a converging lens of focal length f , where would you place the object to obtain an image as far away from the lens as possible? How large would the image be?

Answer: According to the equation $d_i = \frac{d_o f}{d_o - f}$, as $\lim_{d_o \rightarrow f^+} d_i$, the denominator approaches zero, causing d_i to approach infinity. Therefore, placing the object at f results in an image at infinity and the magnification becomes infinitely large.

The experimental results shows a strong correlation with the theory, as evidenced by the low percentage error of %3.2 in the focal length calculation. The data shows that as the object distance d_o decreases, the image distance d_i increases proportionally to satisfy the $1/f$ constant. However, during d_o approached the focal length ($f = 75$ mm), the image was getting harder to focus on the screen, leading to higher errors in magnification measurements (an average error of %15.4). These errors are caused by the difficulty in identifying the place where the most clear image is seen and errors during height measurements. As expected, when $d_o < f$, the lens produced virtual images and we cannot see any image on viewing screen.

Part B: The Telescope

1 Introduction

Telescopes are used to obtain magnified images of distant objects. The image of a distant object when viewed through a single converging lens will be focused nearly at the focal point of the lens. This image will be real, inverted, and reduced in size. In fact, the greater the distance of the object (with respect to f), the smaller the size of the image. However, this reduced image is useful. By viewing this image through a second converging lens—used as a magnifier—an enlarged image can be seen.

2 Theory

Figure 3 shows the setup for a simple telescope. The objective lens, L_1 , creates a real, inverted image. (You can barely see this image in the diagram. It's very small, just inside the focal point of lens L_2 .) If the object is sufficiently far away, this image will be located approximately at f_1 , the focal point of L_1 . The eyepiece, L_2 , then acts as a magnifier, creating a magnified, virtual image which can be viewed by the observer. For maximum magnification, L_2 is positioned so the virtual image is just slightly closer than its focal point, f_2 . Therefore, the distance between the objective lens and the eyepiece of a telescope, when viewing distant objects, is approximately $f_1 + f_2$.

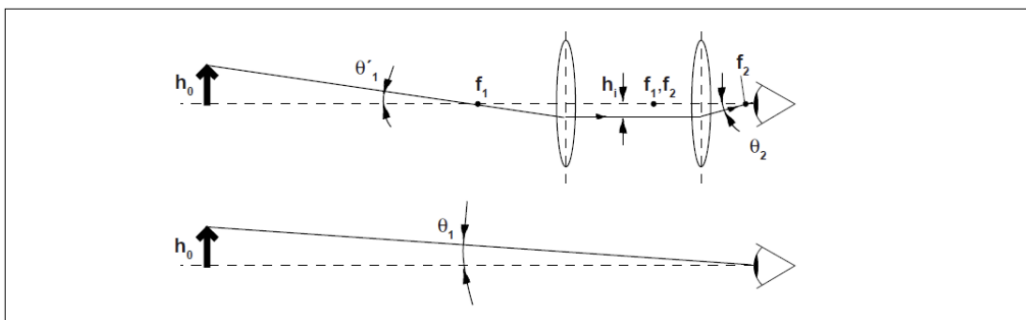


Figure 2: Telescope Magnification

The angular magnification for a telescope can be approximated by assuming the lenses are exactly $f_1 + f_2$ apart, as shown in Figure 2. The height of the

object as seen with the naked eye is proportional to the angle θ_1 in the lower diagram. If the distance from the object to the telescope is large—much larger than is shown in the diagram— $\theta_1 = \theta'_1$, to a good approximation. The ray shown in the upper diagram passes through the focal point of the objective lens, comes out parallel to the optical axis of the telescope, and is therefore refracted by the eyepiece through the focal point of the eyepiece. The angle θ_2 is therefore proportional to h_i , the height of the image seen by the observer.

3 Experimental Setup

Equipment

- Optics Bench
- 75 mm Focal Length Convex Lens
- 150 mm Focal Length Convex Lens
- Component Holders

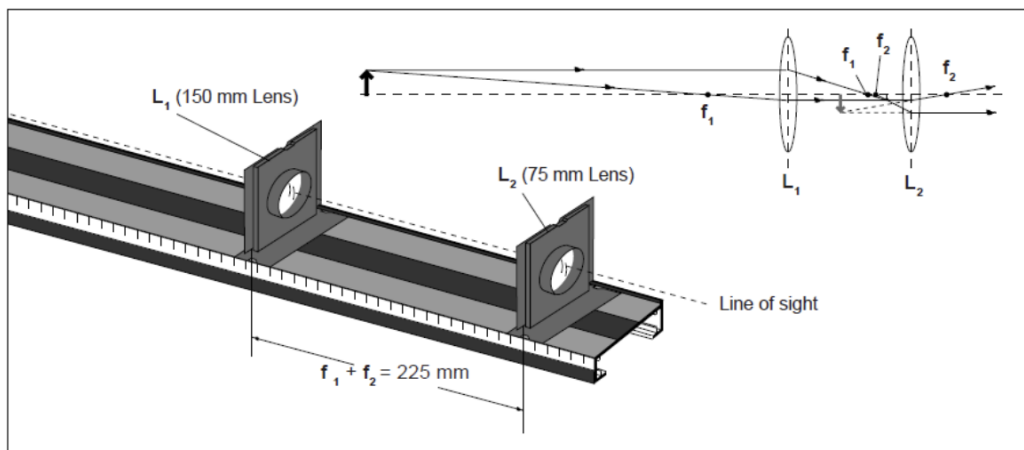


Figure 3: Equipment Setup

Procedure

Set up a telescope using the 75 mm and 150 mm focal length lenses; the distance between the lenses should be approximately 225 mm. Using the 75

mm lens as the eyepiece, look at some reasonably distant object. Adjust the distance between the lenses as needed to bring the object into sharp focus.

To measure the magnification, look with one eye through the telescope, and with the other eye look directly at the object. Compare the size of the two images. (If a meter stick is used as the object, fairly accurate measurements of magnification can be made.)

4 Discussion

4.1 Questions

1. Using Figure 3, calculate $\tan \theta_1$ and $\tan \theta_2$ as a function of the image height h_i and the focal lengths of the two lenses f_1 and f_2 .

Answer:

$$\tan \theta_1 = \tan \theta'_1 = \frac{h_i}{f_1} \quad \tan \theta_2 = \frac{h_i}{f_2}$$

Assuming that θ_1 and θ_2 are very small, and therefore equal to $\tan \theta_1$ and $\tan \theta_2$, respectively.

2. Calculate the magnification of the telescope.

$$\text{Angular magnification} = M = \frac{\theta_2}{\theta_1}$$

Answer:

$$M = \frac{\tan \theta_2}{\tan \theta_1} = \frac{h_i/f_2}{h_i/f_1} = \frac{f_1}{f_2}$$

3. What is the magnification of the telescope when using the 75 mm lens as the eyepiece?

Answer: It was magnified in experiment. Theoretically, using $f_1 = 150$ mm and $f_2 = 75$ mm:

$$M = \frac{f_1}{f_2} = \frac{150}{75} = 2$$

Therefore, the telescope magnifies $2\times$.

4. What is the magnification of the telescope when using the 150 mm lens as the eyepiece and the 75 mm lens as the objective lens?

Answer: It was reduced in experiment. Theoretically, Using $f_1 = 75$ mm and $f_2 = 150$ mm:

$$M = \frac{f_1}{f_2} = \frac{75}{150} = 0.5$$

It reduces size to half.

5. Are your answers to questions 3 and 4 consistent with your answer to question 2?

Answer: Yes. As I also showed in questions 3 and 4, my observations are parallel to the theory in question 2.

By placing the 150 mm objective lens and 75 mm eyepiece lens at a distance equal to the sum of their focal lengths, we successfully observed a magnified, inverted virtual image. Observations in the laboratory were consistent with the theoretical angular magnification of $M = 2$. The experiment confirmed that the magnification of a telescope is fundamentally determined by the ratio of the objective focal length to the eyepiece focal length.

5 Conclusion

In this experiment, the relationship between object distance, image distance, and focal length for a converging lens was investigated. The measurements were used to test the Fundamental Lens Equation and calculate image magnification.

Additionally, a simple telescope was constructed using two converging lenses to demonstrate angular magnification and the basic working principle of telescopes.

For both experiments, the observations satisfied the theory.